

Let Both Sides Adapt

Parameter-efficient asymmetric brain–vision alignment
for EEG image retrieval

YOUR NAME | YOUR STUDENT ID

YOUR TEAM MEMBERS

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AIAA3800 — Human-Centered Artificial Intelligence
AI Thrust, HKUST (GZ)



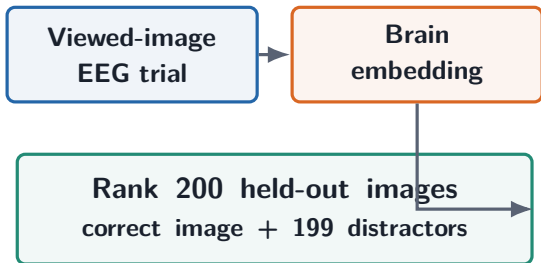
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Can 17 EEG channels recover one image from 200?

THE TASK

A person views an image.
We ask whether their
non-invasive EEG can
retrieve that **exact image**
from a held-out gallery.

17 posterior channels
250 time samples
repeated-trial averaging



Random Top-1

0.5%

Our five-seed Top-1

86.66%

Source: THINGS-EEG2; project fixed-checkpoint 200-way exact-image retrieval protocol.

Freeze the visual encoder

Preserves CLIP's pretrained semantics, but forces the EEG encoder to absorb the entire cross-modal gap.

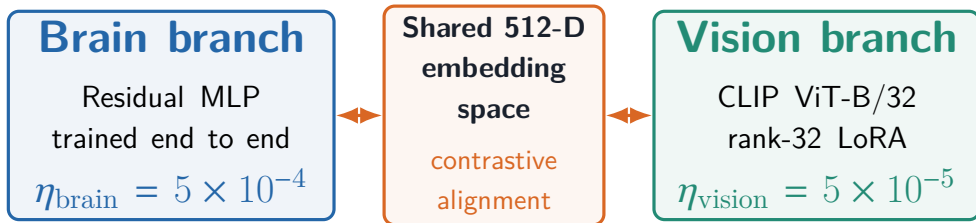
Fully fine-tune the visual encoder

Lets vision move freely, but small EEG datasets make overfitting and loss of the pretrained visual prior more likely.

The two modalities should meet in the middle —

while vision moves less.

OUR METHOD: ASYMMETRIC CO-ADAPTATION



Brain updates are 10× faster than visual updates.

Vision adapts through all-linear LoRA; the pretrained CLIP weights remain frozen.

Source: Brain residual MLP; CLIP ViT-B/32; LoRA rank/alpha 32; branch learning rates 5e-4 and 5e-5.

TRAINING

$$z_i^b = \frac{E_b(b_i)}{\|E_b(b_i)\|_2}, \quad z_j^v = \frac{E_v(x_j)}{\|E_v(x_j)\|_2}$$

$$S_{ij} = \exp(\alpha) (z_i^b)^\top z_j^v$$

Brain-to-image contrastive loss pushes each paired image above the other images in the batch.

EVALUATION

One EEG query

↓ 200 cosine scores ↓

Independent Top-1 /
Top-5 ranking

No one-to-one assignment and no Hungarian decoding in the standard result.

Source: Repository implementation and audited fixed-checkpoint evaluation protocol.

MAIN SYSTEM: 10 SUBJECTS $\times$ 5 SEEDS

86.66%

Top-1 accuracy
 ± 0.69 percentage points

98.38%

Top-5 accuracy
 ± 0.14 percentage points

200-way chance

0.5% / 2.5%

Top-1 seed range

85.60–87.35%

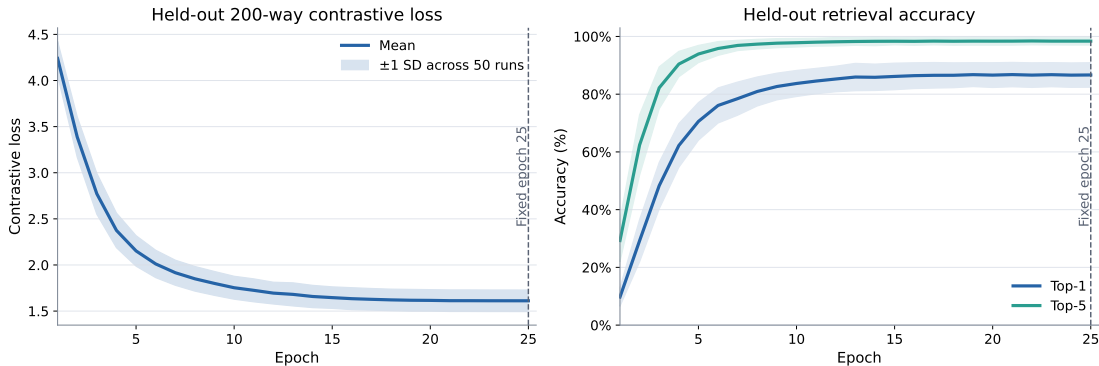
Checkpoint rule

fixed epoch 25

Source: Five seed-level subject macros; 200 held-out queries per subject–seed model.

Fifty independent runs converge to the same operating point

Training Dynamics Across 50 Subject-Seed Runs



Diagnostic trajectories on the THINGS-EEG2 test split; checkpoint fixed at epoch 25, with no epoch selection.

Every reported model uses epoch 25; the curves are diagnostics, not model selection.

Source: 50 subject-seed trajectories: seeds 42–46, subjects 01–10.

Matched control isolates a +6.51-point gain

CONTROLLED ATTRIBUTION — SEPARATE SAMGA-STYLE SYSTEM

Matched visual arm	Top-1	Top-5
Frozen CLIP	76.17%	95.79%
+ visual LoRA/TTUR	82.68%	97.67%

+6.51 points

95% CI: [+5.22, +7.69]

48 / 50 paired cells improved

**All 10 subjects improved on
average**

Source: Matched 10-subject $\times$ 5-seed study; two-way subject/seed cluster bootstrap, 10,000 resamples.

The evidence supports

- ☐ Within-subject retrieval
- ☐ Trial-averaged EEG
- ☐ A closed 200-image gallery
- ☐ Stable five-seed performance

It does not support

- ☐ Cross-subject generalization
- ☐ Single-trial “mind reading”
- ☐ Open-world reconstruction
- ☐ Clinical or diagnostic use

**Consent, privacy, fixed evaluation rules, and restrained claims
are part of the model’s human-centered quality.**

Source: Project limitations, evaluation policy, and responsible-use statement.

Do not force EEG into a rigid visual space.

Let EEG learn fast, while vision adapts slowly.

Asymmetric

Brain updates 10× faster than vision.

Efficient

Rank-32 LoRA preserves the CLIP prior.

Supported

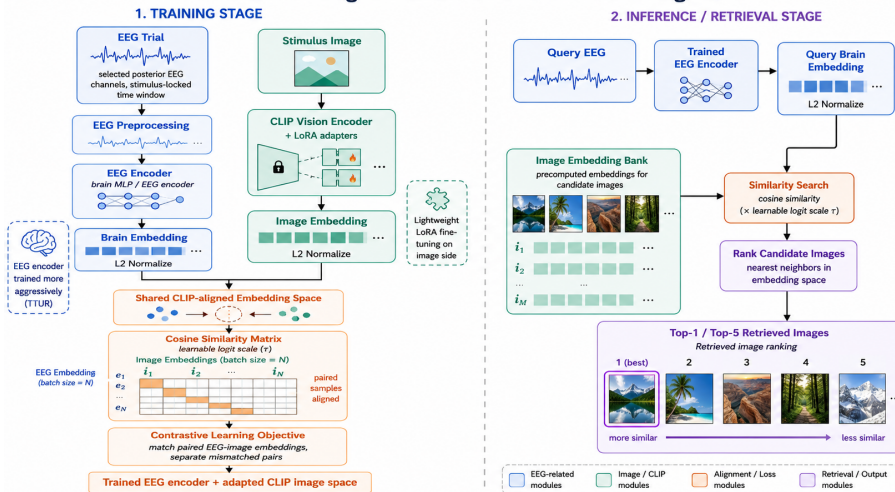
Strong main result and matched-control gain.

Next: test whether the principle survives cross-subject and single-trial settings.

Thank you!

Questions?

EEG-to-Image Retrieval via Dual-Side CLIP Alignment



Source: Project-authored architecture visual.

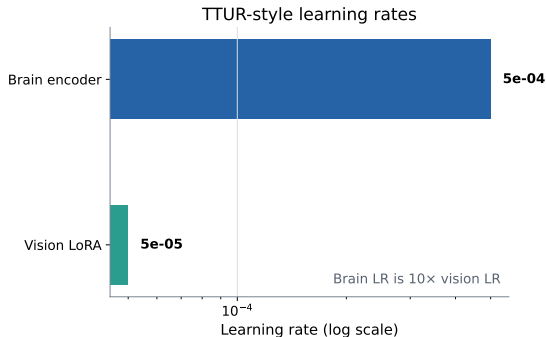
Image Encoder LoRA — Verified Final Configuration (No Local Hyperparameter Sweep Recorded)

Parameter-efficient vision adaptation

Frozen CLIP ViT-B/32 weights

Trainable LoRA adapters · rank=32 · alpha=32

All-linear targets · q/k/v/out · fc1/fc2
· visual projection



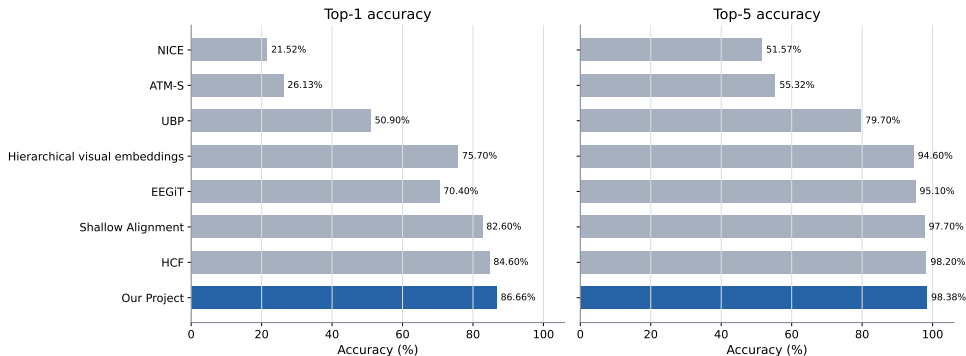
LoRA dropout=0.0 · cosine scheduler · weight decay 0.05 · 25 epochs · fixed final checkpoint

This documents the final configuration; no local sweep supports an optimality claim.

Source: Verified project configuration; rank/alpha 32, dropout 0, cosine schedule, weight decay 0.05, 25 epochs.

Protocols differ; **SAMGA** is intentionally omitted from this plot; no unqualified **SOTA** claim.

Standard 200-way EEG-to-Image Retrieval — Literature Comparison

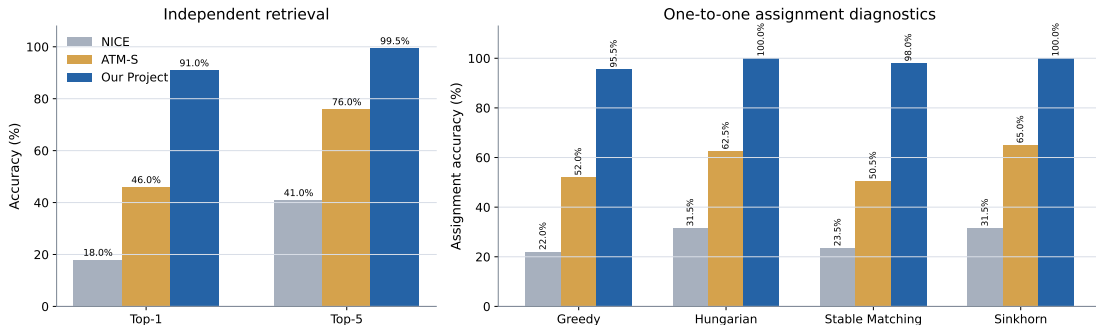


Headline cross-paper values; visual targets, encoders, schedules, and checkpoint rules differ.

Source: Headline cross-paper values; visual targets, encoders, schedules, and checkpoint rules differ.

Subject-08, seed 42 only; global methods inspect the complete similarity matrix.

Same-Query, Same-Gallery Comparison · Subject-08 · Seed 42 · 200-way



Global assignment methods inspect the complete similarity matrix and are transductive diagnostics, not standard retrieval.

Source: Same-query, same-gallery local comparison; Hungarian is excluded from every standard aggregate.

Track	Correct interpretation
Main system	Fixed epoch-25 independent retrieval: 86.66% / 98.38%.
Controlled LoRA	Matched Frozen vs. LoRA/TTUR test in a SAMGA-style system.
SAMGA reproduction	Audited approximation; separate from our main method.
Hungarian	One Subject-08/seed-42 transductive diagnostic.
Reconstruction	Exploratory extension without a formal headline result.

Source: Repository result taxonomy and claim boundaries.

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- [2] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [3] L. Jiang, Q. She, J. Xu, H. Xu, D. Wu, and Z. Kuang. Subject-aware multi-granularity alignment for zero-shot eeg-to-image retrieval. *arXiv preprint arXiv:2604.17782*, 2026.
- [4] D. Li, C. Wei, S. Li, J. Zou, and Q. Liu. Visual decoding and reconstruction via eeg embeddings with guided diffusion. In *Advances in Neural Information Processing Systems*, 2024.
- [5] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.